(a) DHW-conditioned future severe rate
(week-block bootstrap 95% CI)

SCS
GBR
CARIB

future severe rate P[y(t+4) ≥ 0.8]

current DHW bin center (°C-weeks)

(b) Basin conditional-risk difference at matched DHW
(CI excl. 0 = basin-specific)

CARIB − GBR
CARIB − SCS
GBR − SCS

risk difference at matched DHW

current DHW bin center (°C-weeks)

(c) Residual R tertiles within matched DHW bins
(light→dark = R Q1→Q3; bold = Q3)

SCS R-Q1
SCS R-Q2
SCS R-Q3
GBR R-Q2
GBR R-Q3
CARIB R-Q2
CARIB R-Q3

future severe rate within DHW bin

current DHW bin center (°C-weeks)

(d) DHW-only (A) vs DHW+R (B) — matched-strata gain
solid bold = B; dashed = A; dotted = TSAR reference

SCS: ΔROC -0.0110 [-0.0174, -0.0039]
GBR: ΔROC +0.0063 [-0.0032, +0.0155]
CARIB: ΔROC -0.0059 [-0.0173, +0.0032]

SCS A (DHW) AUC 0.980 / AP 0.235
SCS B (DHW+R) AUC 0.969 / AP 0.155
SCS TSAR (ref) AUC 0.979 / AP 0.227
GBR A (DHW) AUC 0.953 / AP 0.282
GBR B (DHW+R) AUC 0.960 / AP 0.393
GBR TSAR (ref) AUC 0.953 / AP 0.286
CARIB A (DHW) AUC 0.975 / AP 0.311
CARIB B (DHW+R) AUC 0.969 / AP 0.507
CARIB TSAR (ref) AUC 0.979 / AP 0.368

True positive rate

False positive rate